

CSC 242: DISCRETE STRUCTURE


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HEAD OF DEPARTMENT
Date: *20/2/24*

B.Sc. EXAMINATION

COLLEGE: SCIENCE & TECHNOLOGY **DEPARTMENT: COMPUTER & INFO. SCIENCES**
SESSION: 2023/2024 **SEMESTER: ALPHA**
COURSE CODE: CSC416 **CREDIT UNIT: TWO (2)**
COURSE TITLE: DISCRETE STRUCTURES
INSTRUCTION: ATTEMPT ANY THREE (3) QUESTIONS **TIME: 2 HOURS**

1. (a) Consider the following statements:
"All hummingbirds are richly colored."
"No large birds live on honey."
"Birds that do not live on honey are dull in color."
"Hummingbirds are small."
Let $P(x)$, $Q(x)$, $R(x)$, and $S(x)$ be the statements "x is a hummingbird", "x is large", "x lives on honey", and "x is richly colored", respectively. Assuming that the domain consists of all birds, express the statements in the argument using quantifiers and $P(x)$, $Q(x)$, $R(x)$, and $S(x)$. (8 marks)

(b) How many different elements does $A \times B \times C$ have, if A has m elements, B has n elements, and C has p elements? (2½ marks)

(c) Find A^3 , if
(i) $A = \{a\}$ (3 marks)
(ii) $A = \{0, a\}$ (3 marks)

(d) (i) State the pigeonhole principle. (2 marks)
(ii) Let (x_i, y_i) , $i = 1, 2, 3, 4, 5$ be a set of five distinct points with integer coordinates in the xy plane. Show that the midpoint of the line joining at least one pair of these points has integer coordinates. (5 marks)

2. (a) Soccer matches between two teams consists of at most five (5) games. The first team that wins three games wins the playoff.
(i) Construct a tree diagram for five games. (5 marks)
(ii) In how many different ways can the playoff occur? (3 marks)

(b) Highlight four real-life scenarios where graph models are applicable, with two relevant examples each. (8 marks)

(c) Highlight the various key terminologies of a Rooted Tree. Support your answer with a diagrammatical illustration. (7½ marks)

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3. (a) Suppose that a universal set, $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Express the following sets with bit strings where the i^{th} bit in the string is 1 if i is in the set and 0 otherwise.

(i) $\{3, 4, 5\}$

(1 marks)

(ii) $\{1, 3, 6, 10\}$

(1 marks)

(iii) $\{2, 3, 4, 7, 8, 9\}$

(1 marks)

(b) Write a C++ program that dynamically takes these sets in 3(i-iii) above as input and displays the corresponding bit strings for each set. (7 marks)

(c) Find $f \circ g$ and $g \circ f$, where $f(x) = x^2 + 1$ and $g(x) = x + 2$, are functions from R to R .

State if $f \circ g = g \circ f$.

(3½ marks)

(d) Let A be the set $\{1, 2, 3, 4\}$. Consider the following relations on A :

$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$,

$R_2 = \{(1, 1), (1, 2), (2, 1)\}$,

$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$,

$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$,

$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$,

$R_6 = \{(3, 4)\}$.

Determine the following:

(i) Which ordered pairs are in the relation $R = \{(a, b) \mid a \text{ divides } b\}$?

(2 marks)

(ii) Which of these relations are reflexive?

(2 marks)

(iii) Which of these relations are symmetric?

(2 marks)

(iv) Which of these relations are antisymmetric?

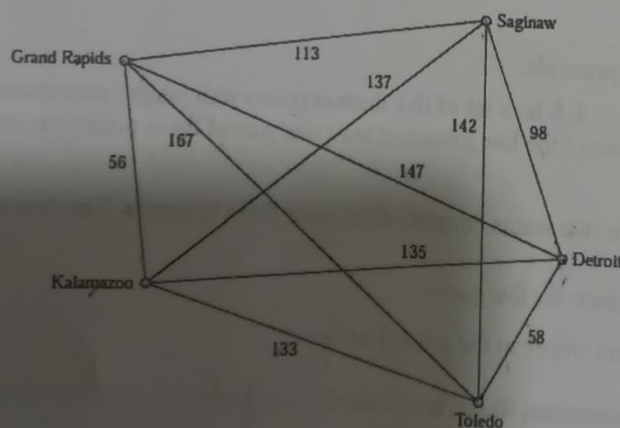
(2 marks)

(v) Which of these relations are Transitive?

(2 marks)

(a) Explain in details (and with diagrammatic illustration) the types of undirected graph. (9 marks)

(b) A Christian missionary is to embark on a trip from the city of Detroit through all other cities and returns to Detroit, in which order should he visit these cities to travel the minimum distance?



(10 marks)

Highlight, with a relevant example each, three application domains of Tree models in computing.

(4½ marks)

3. (a) Define the Handshaking Theorem.

(3 marks)

(b) An Airplane pilot has been scheduled for a trip to five (5) different cities. He can travel directly between two cities through the same route.

(i) Can he visit exactly three (3) cities from each of the cities on the flight plan? Compute your answer.

(3 marks)

(ii) Support your answer with a diagrammatical illustration.

(2 marks)

(c) In the Graph diagram below, present in tabular form the following:

(i) Identify each vertex and state the total number of vertices.

(1 marks)

(ii) Identify each edge per vertex and state total number of edges.

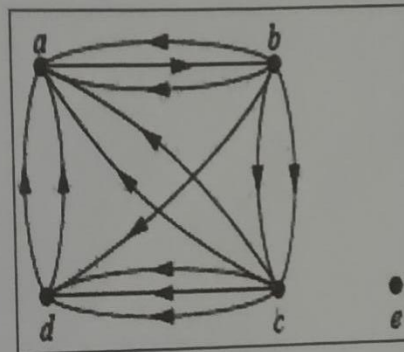
(1 marks)

(iii) Identify each vertex's in-degree and state the total number of the in-degrees.

(2 marks)

(iv) Identify each vertex's out-degree and state the total number of the out-degrees.

(2 marks)



(d) Suppose CU sends out convocation invitation cards to graduating students, each student who receives the cards is asked to send them on to four (4) other students. While some students did send out the cards, others did not. If no one receives more than one card and if the card circulation ends after there have been 100 people who read it but did not send it out.

(i) How many people have seen the card, including the first person?

(6 marks)

(ii) How many people sent out the card?

(3½ marks)